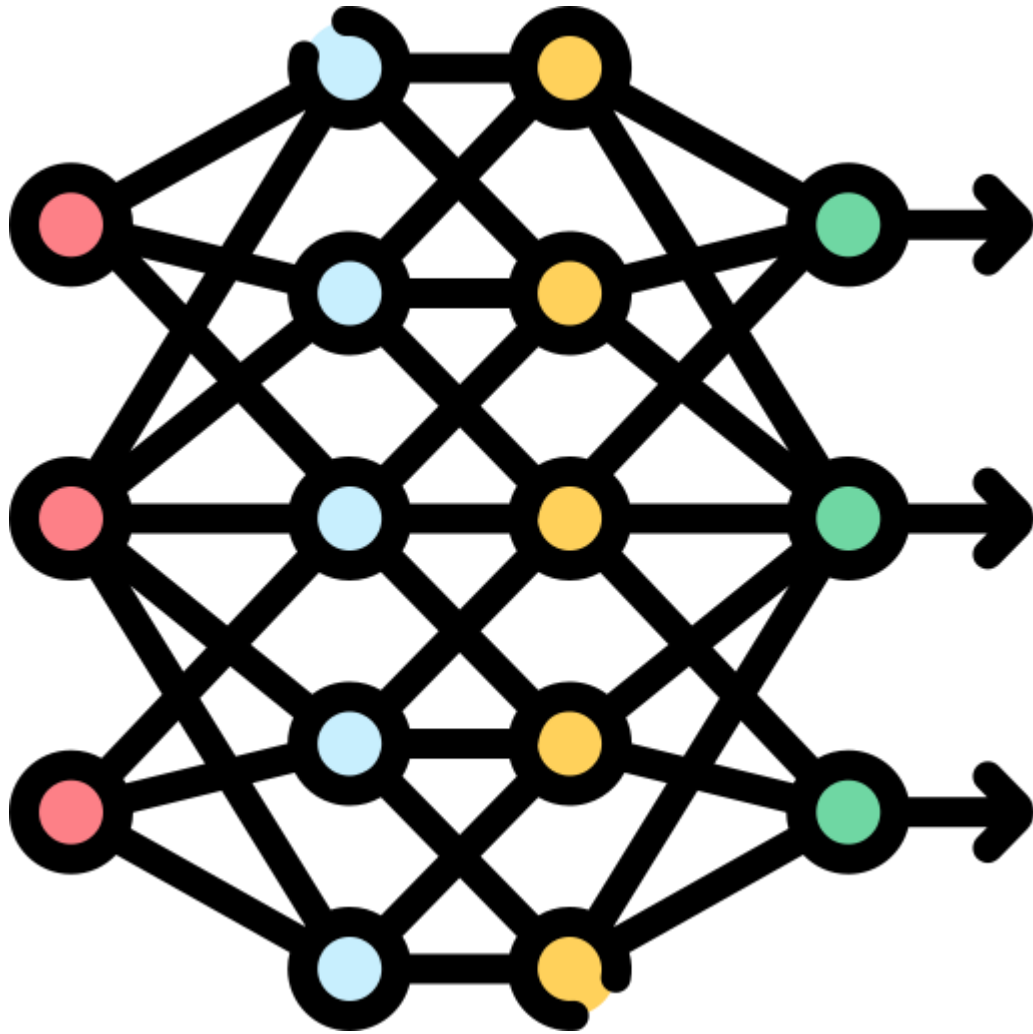


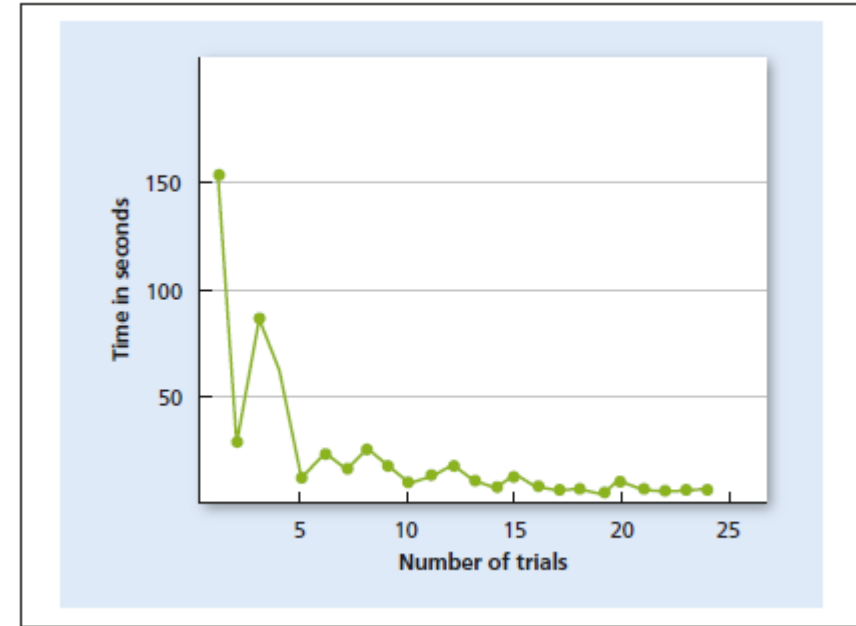


LEARNING

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Theories of Learning: Operant Conditioning

- Classical conditioning - learning of involuntary, automatic responses
- Operant conditioning - organisms learning of voluntary responses
- Edward L. Thorndike and B. F. Skinner
- Experiment of the 'puzzle box' by Thorndike
 - Trial-and-error learning



- Hungry cat placed in puzzle box – press lever to escape
- Trial and error
- Cat pushes the lever and opens the door
- Lever (stimulus), pushing the lever (response), consequence-escaping (good) and food (better)



Theories of Learning: Operant Conditioning (cont'd...)

Law of effect:

- If an action is followed by a pleasurable consequence, it will tend to be repeated
- If an action is followed by an unpleasant consequence, it will tend not to be repeated

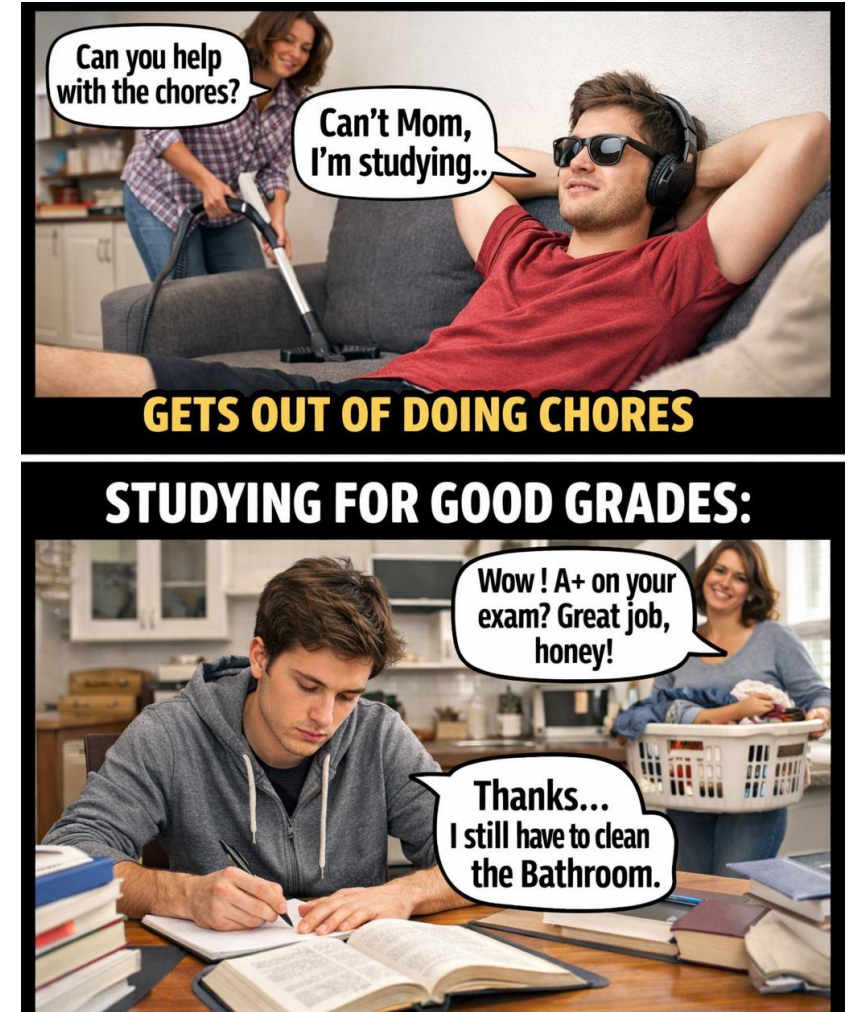
Operant conditioning:

- The learning of voluntary behavior through the effects of pleasant and unpleasant consequences to responses
- Operant = any behavior that is **voluntary** and not elicited by specific stimuli
- B. F. Skinner
- Effect of consequences on behavior
- E.g., “If I do this, what’s in it for me?”

Theories of Learning: Operant Conditioning (cont'd...)

“What’s in it for me?”

- **Reinforcement:** Any event or stimulus, that when following a response, increases the probability that the response will occur again
- “To strengthen”
- Related to the law of effect
- E.g., work and paycheck (pleasant consequence); avoiding doing chores (unpleasant consequence)



Theories of Learning: Operant Conditioning (cont'd...)

Types of Reinforcement

Primary reinforcer:

Any reinforcer that is naturally reinforcing by meeting a basic biological need, such as hunger, thirst, or touch

Positive reinforcement:

The reinforcement of a response by the addition or experiencing of a pleasurable stimulus

E.g., getting a shout-out on the university's social media page after volunteering at a campus event

Secondary reinforcer:

Any reinforcer that becomes reinforcing after being paired with a primary reinforcer, such as praise, tokens, or gold stars

Negative reinforcement:

The reinforcement of a response by the removal, escape from, or avoidance of an unpleasant stimulus

E.g., the professor stops sending reminder emails about missed assignments once a student begins submitting all work on time

BOTH positive AND negative reinforcement *increase* the likelihood of the behavior they follow—they *both* have the effect of strengthening, or reinforcing, the behavior

Theories of Learning: Operant Conditioning (cont'd...)

Schedule Type	Definition	Example
Fixed Ratio	Reinforcement after a <u>set number of responses</u>	Student receives a reward for every 5 questions answered correctly on an online quiz
Variable Ratio	Reinforcement after an <u>unpredictable number of responses</u>	Extra credit is given after a random number of class participation points- could be the 2nd, 7th, or 10th time a student chimes in Slot machines or games
Fixed Interval	Reinforcement after a <u>fixed amount of time passes</u>	Weekly review session: Students get bonus points if they submit weekly reflection papers on time Salary
Variable Interval	Reinforcement after a varying, <u>unpredictable amount of time</u>	Surprise pop quizzes are given; students never know exactly when, sometimes after 3 days, sometimes after 8 days

Theories of Learning: Operant Conditioning (cont'd...)

When you want to weaken the response

Punishment by Application

Something unpleasant is added to the situation to reduce the response

For example: Scolding, extra homework,

Punishment by removal

Behaviour is punished by removal of something pleasurable or desired after the behaviour occurs

For example: Removing freedom by grounding, taking away phone (going out with friends or talking on phone)

	Reinforcement	Punishment
Positive (Adding)	Positive Reinforcement Something valuable or desirable is added	Punishment by Application Something unpleasant is added
Negative (Removing/ avoiding)	Negative reinforcement Something unpleasant is removed	Punishment by removal Something desirable is removed

Theories of Learning: Operant Conditioning (cont'd...)

Which of the following examples would be reinforcement or punishment and which would be negative reinforcement:

1. P's father nags him to wash his car. P hates being nagged, so he washes the car so his father will stop nagging
2. N learns that talking in a funny voice gets him lots of attention from his classmates, so now he talks that way more often
3. A student keeps talking during lectures. The teacher responds by assigning extra homework each time the student talks. Over time, the student talks less in class
4. R repeatedly watches television and procrastinating. After several warnings, her mother takes away her television privileges for three days. After this, R stops procrastinating and is doing her homework

1. Negative reinforcement: P is being negatively reinforced for washing his car because the nagging (unpleasant stimulus) stops when he does so
2. Positive reinforcement: N is getting positive reinforcement in the form of his classmates' attention
3. Punishment by application: Student is punished by adding extra homework hence, student talks less
4. Punishment by removal: TV privileges were removed leading reducing procrastination

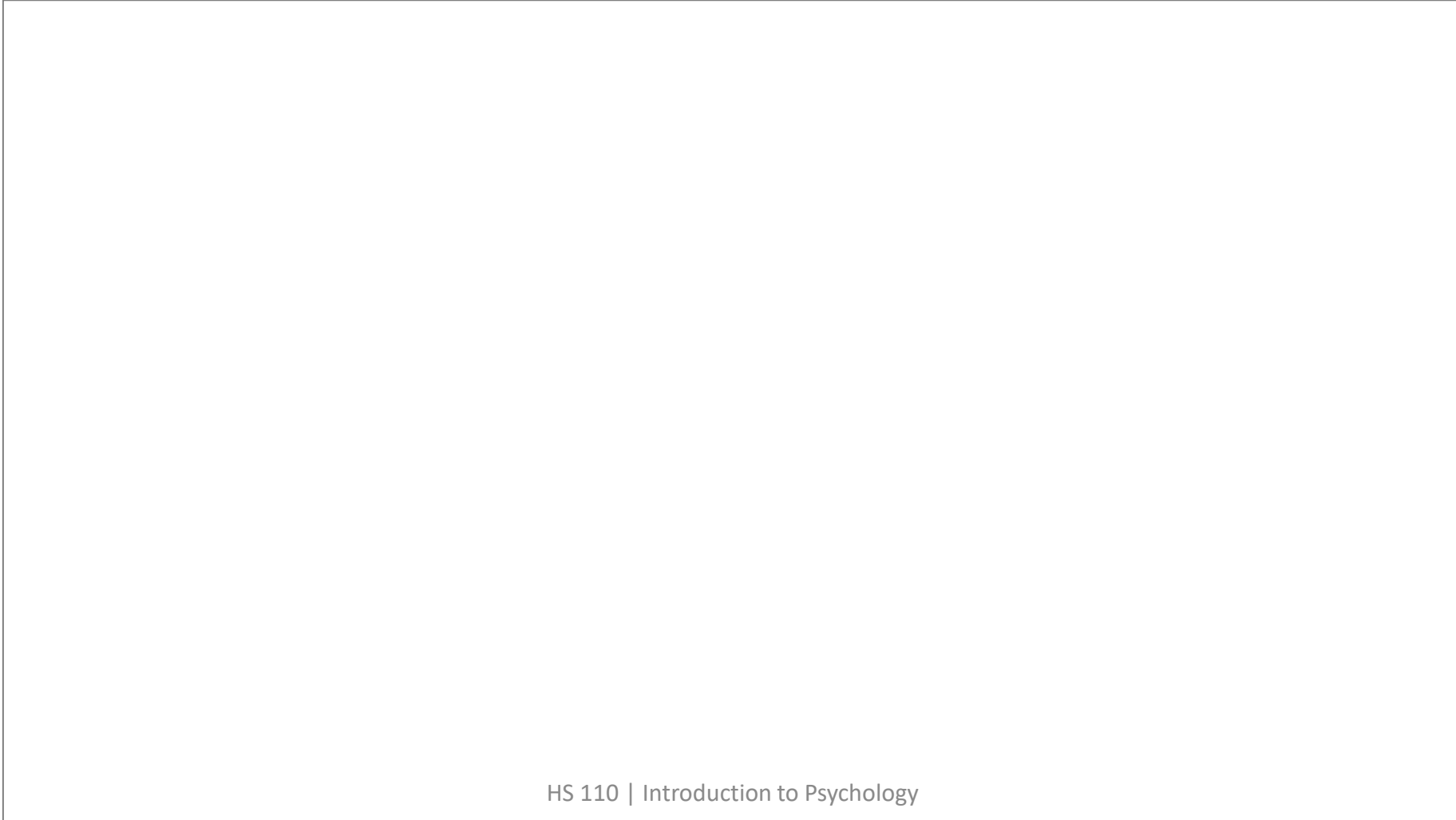
What type of reinforcement or punishment worked best for you when you were in school?

Theories of Learning: Operant Conditioning (cont'd...)



Theories of Learning: Observational Learning

Learning new behavior by watching a model perform that behavior



Theories of Learning: Observational Learning (cont'd...)

Four elements of observational learning:

Attention: To learn anything through observation, the learner must first pay *attention* to the model

E.g., watching a senior student answer questions confidently in discussion

Memory: The learner must also be able to retain the *memory* of what was done

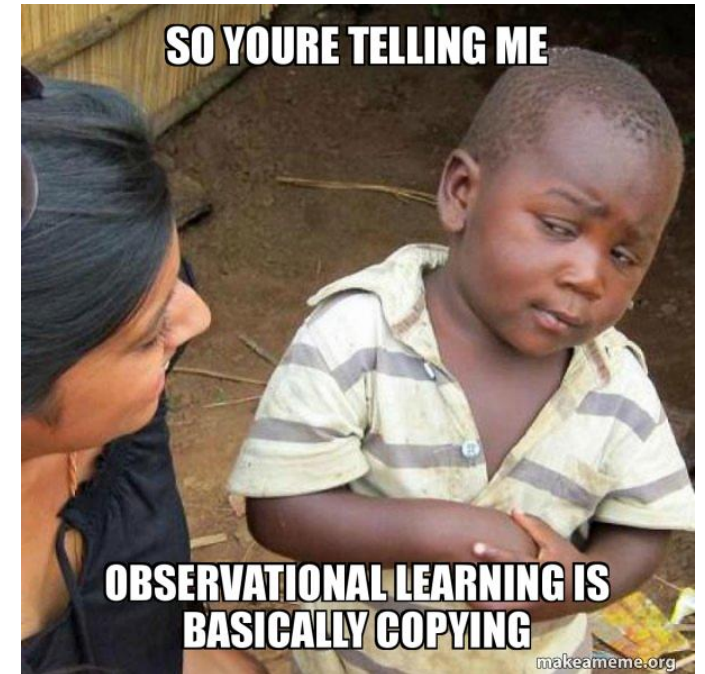
E.g., remembering the key arguments and style used

Imitation: The learner must be capable of reproducing, or *imitating*, the actions of the model

E.g., using a similar approach in their own discussion response

Desire: Finally, the learner must have the desire or *motivation* to perform the action

E.g., driven by wanting peer approval and good participation marks



Conclusion

- Operant conditioning
 - Positive reinforcement
 - Negative reinforcement
 - Punishment by application
 - Punishment by removal
- Observational learning
 - Attention
 - Memory
 - Imitation
 - Desire